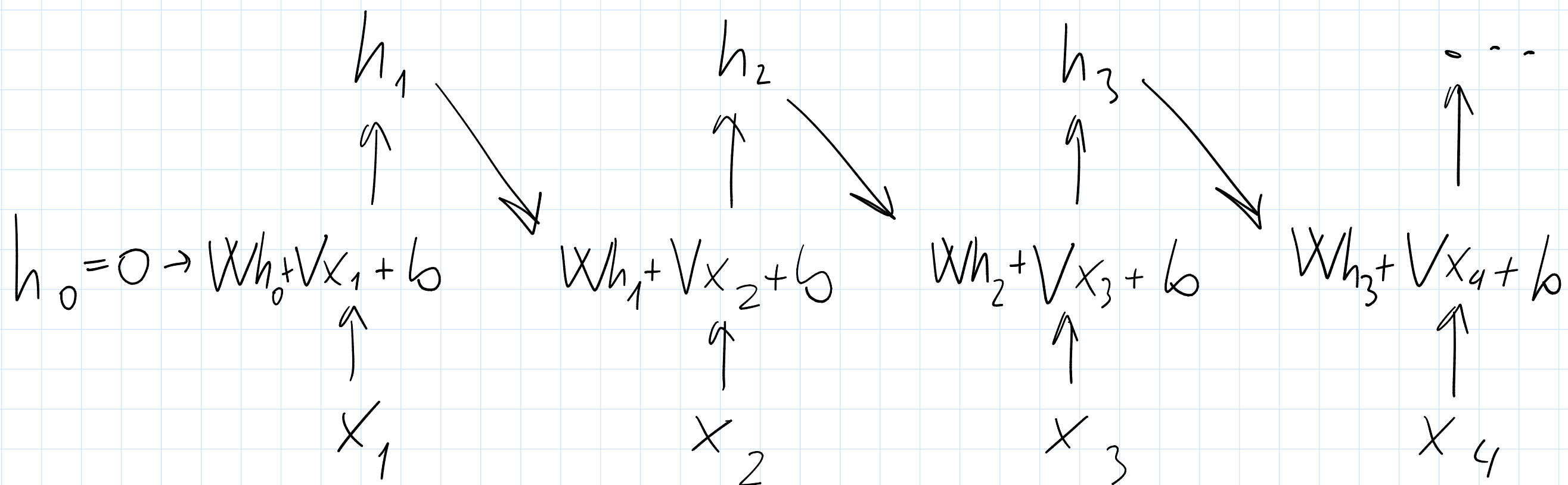


Recurrent Neural Networks

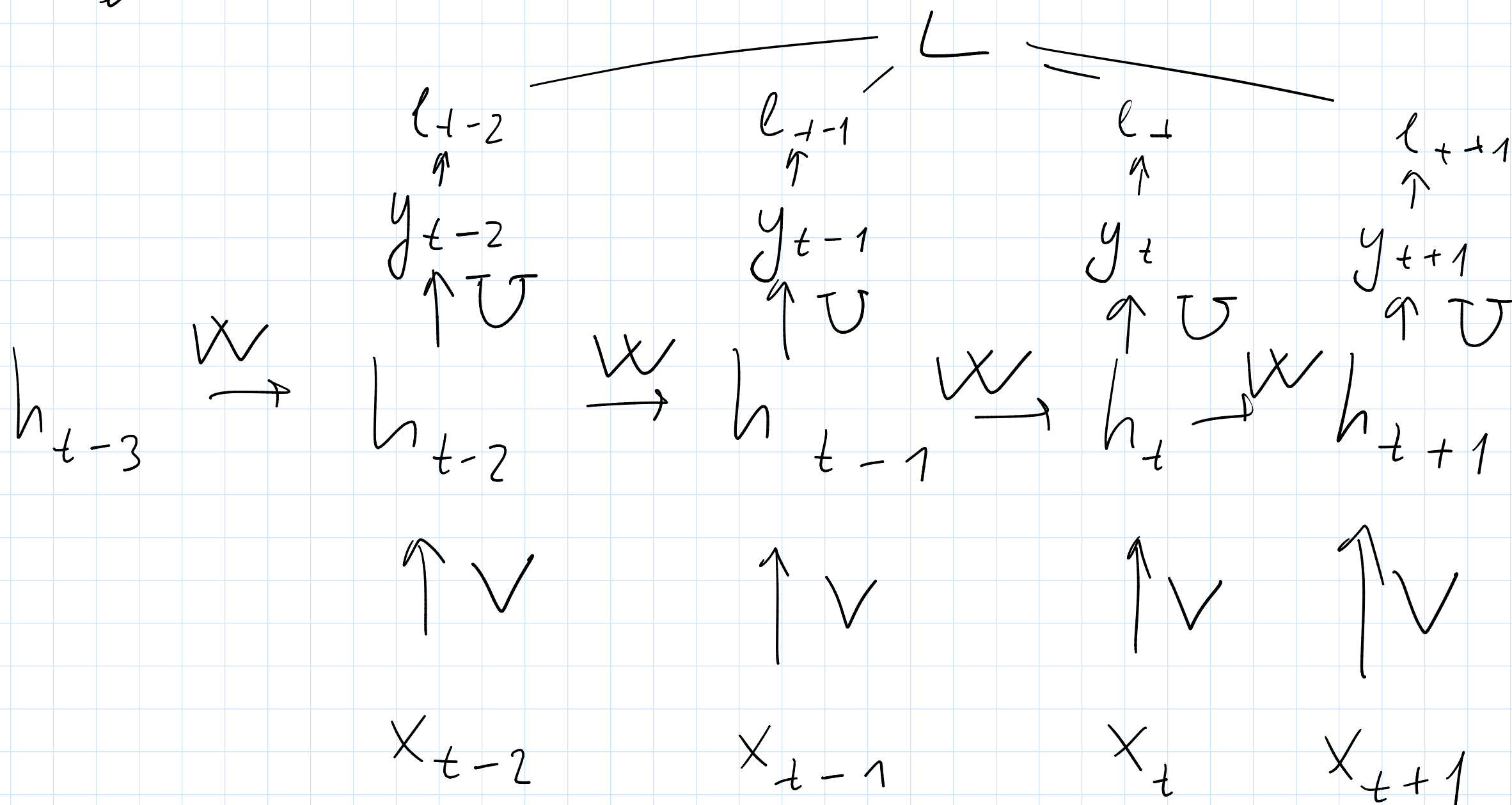


$$x_i \in \mathbb{R}^d$$

$$\begin{pmatrix} V & | & W \end{pmatrix} \begin{pmatrix} x_t \\ h_{t-1} \end{pmatrix}$$

h - hidden state

$$h_t = f(Wh_{t-1} + Vx_t + b)$$



$$L = \frac{1}{T} \sum_{t=1}^T \ell_t$$

$$\frac{dL}{dU} = \frac{1}{T} \sum_{t=1}^T \frac{d\ell_t}{dU} = \frac{1}{T} \sum_{t=1}^T \frac{d\ell_t}{dy_t} \frac{dy_t}{dU}$$

$$\frac{dL}{dW} = \frac{1}{T} \sum_{t=1}^T \frac{dL_t}{dW} = \frac{1}{T} \sum_{t=1}^T \frac{dL_t}{dy_t} \cdot \frac{dy_t}{dh_t} \cdot \frac{dh_t}{dW}$$

$$\frac{dh_t}{dW} = \frac{\partial h_t}{\partial W} + \frac{\partial h_t}{\partial h_{t-1}} \cdot \frac{dh_{t-1}}{dW} = \frac{\partial h_t}{\partial W} + \frac{\partial h_t}{\partial h_{t-1}} \cdot \frac{\partial h_{t-1}}{\partial W} +$$

$$+ \frac{\partial h_t}{\partial h_{t-1}} \cdot \frac{\partial h_{t-1}}{\partial h_{t-2}} \cdot \frac{dh_{t-2}}{dW} = \sum_{j=0}^{t-1} \left(\prod_{k=t-j}^{t-1} \frac{\partial h_{k+1}}{\partial h_k} \right) \frac{\partial h_{t-j}}{\partial W}$$

$$\frac{dL}{dW} = \frac{1}{T} \sum_{t=1}^T \frac{dL_t}{dy_t} \frac{dy_t}{dh_t} \sum_{j=0}^{t-1} \left(\prod_{k=t-j}^{t-1} \frac{\partial h_{k+1}}{\partial h_k} \right) \frac{\partial h_{t-j}}{\partial W}$$

BPTT (Backpropagation through time)

$$\prod \frac{\partial h_{k+1}}{\partial h_k}$$

$$\bullet \left\| \frac{\partial h_{k+1}}{\partial h_k} \right\| > 1 \quad - \text{exploding gradients, divergence, NaNs}$$

Gradient clipping

$$g \leftarrow \min \left(1, \frac{\text{thr}}{\|grad\|} \right) \cdot grad$$

$$g \leftarrow \text{clip}(grad, c, -c)$$

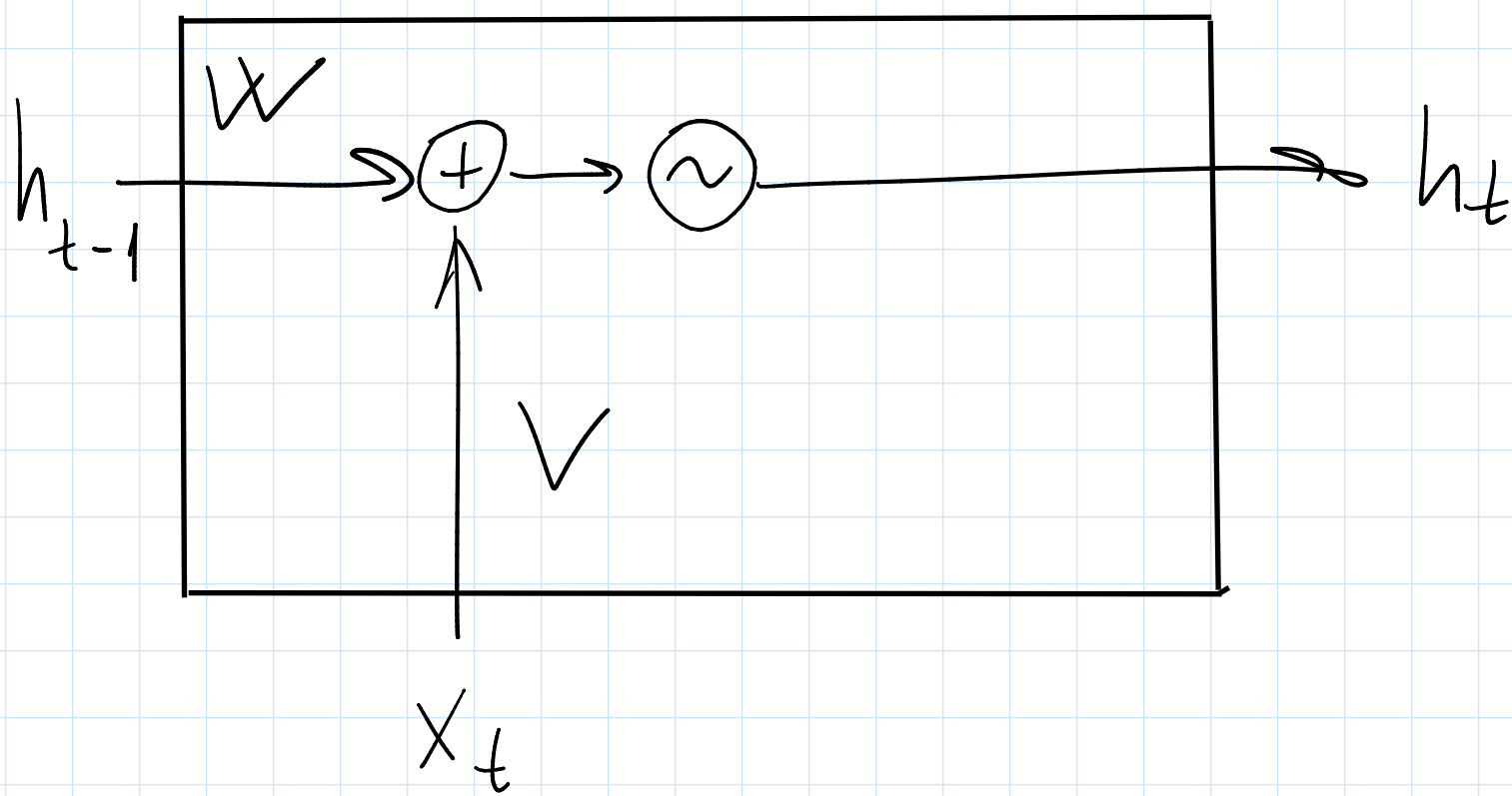
$$\bullet \left\| \frac{\partial h_{k+1}}{\partial h_k} \right\| < 1 \quad - \text{vanishing gradients}$$

$$h_{k+1} = f(\underbrace{W h_k + V x_{k+1} + b}_{z_{k+1}})$$

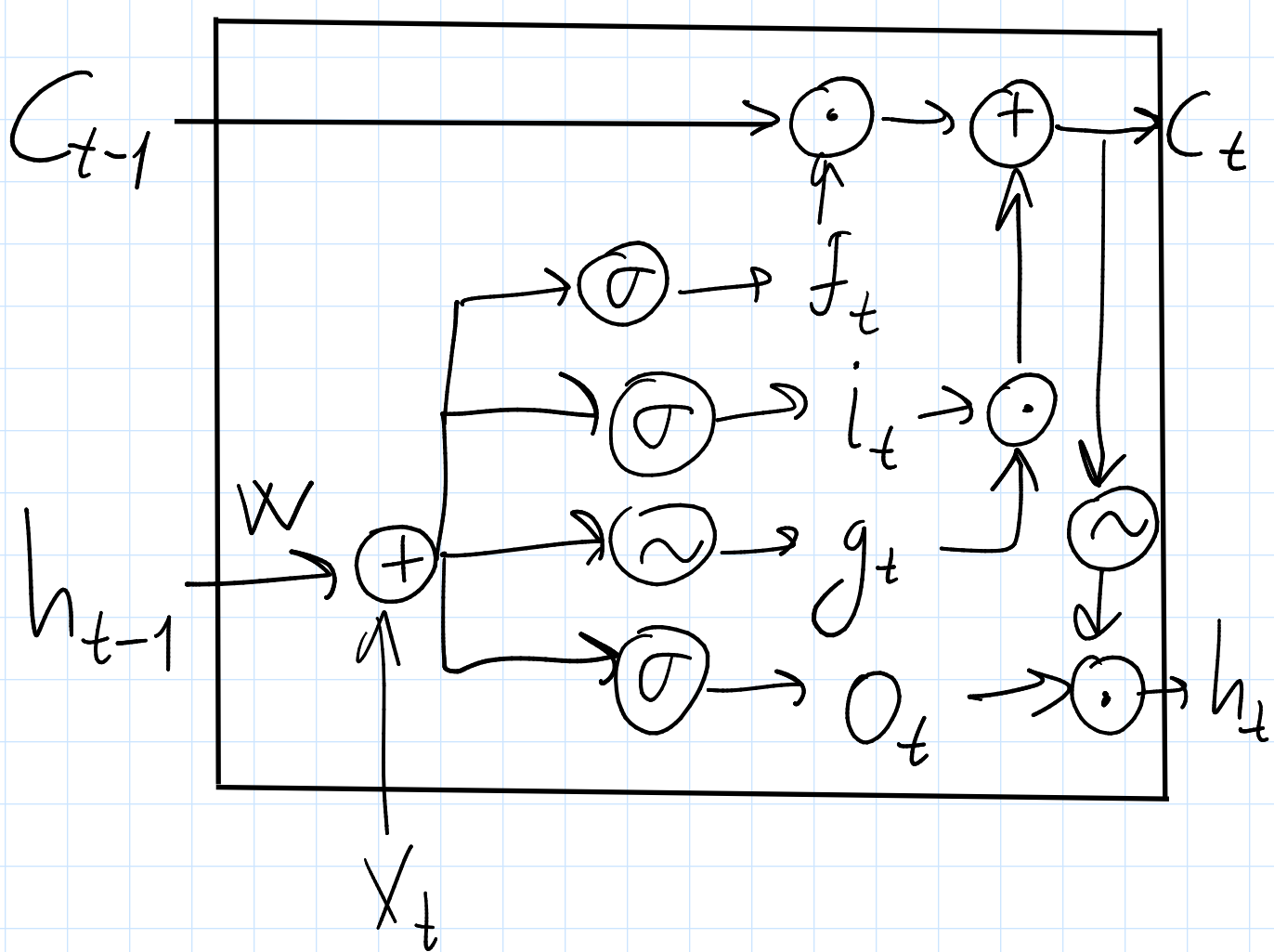
Init W with orthogonal matrix

$$\left\| \frac{\partial h_{k+1}}{\partial h_k} \right\| = \left\| \text{diag} \left(f'_{h_k}(z_{k+1}) \right) \right\| \cdot \|W\|$$

RNN



LSTM (long short term memory)



$$g_t = f(W_g h_{t-1} + V_g x_t + b_g)$$

$$i_t = \sigma(W_i h_{t-1} + V_i x_t + b_i)$$

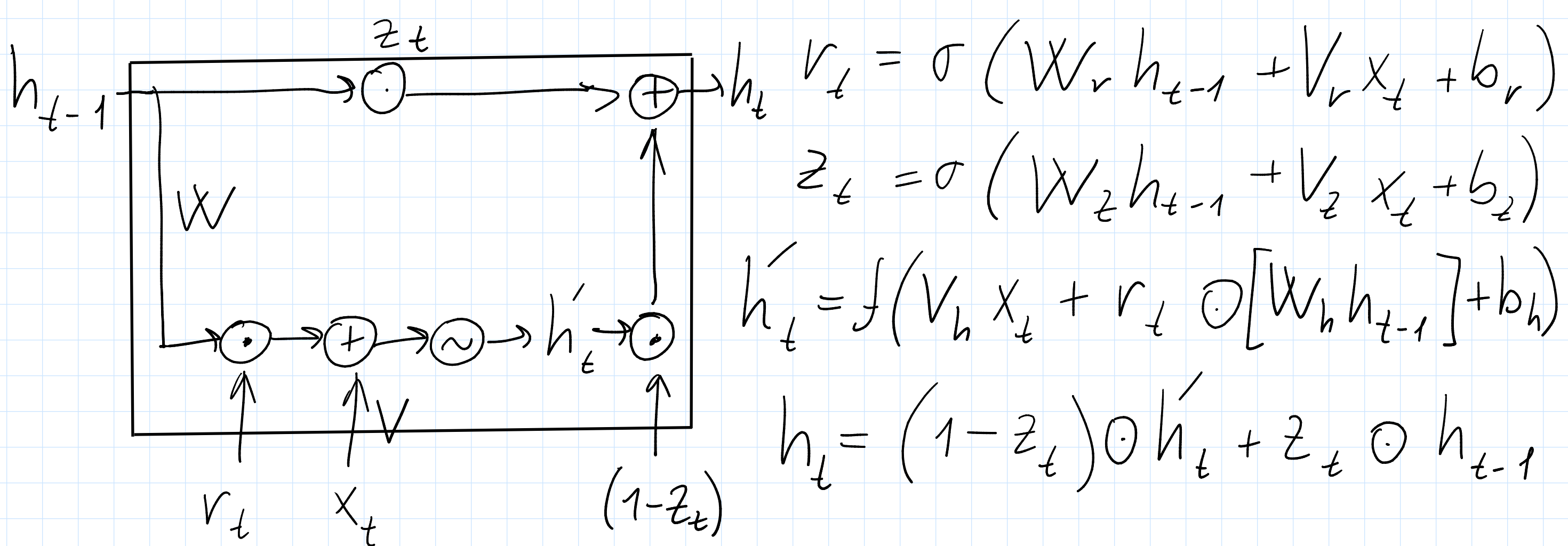
$$o_t = \sigma(W_o h_{t-1} + V_o x_t + b_o)$$

$$f_t = \sigma(W_f h_{t-1} + V_f x_t + b_f)$$

$$c_t = c_{t-1} \odot f_t + i_t \odot g_t$$

$$h_t = f(c_t) \odot o_t$$

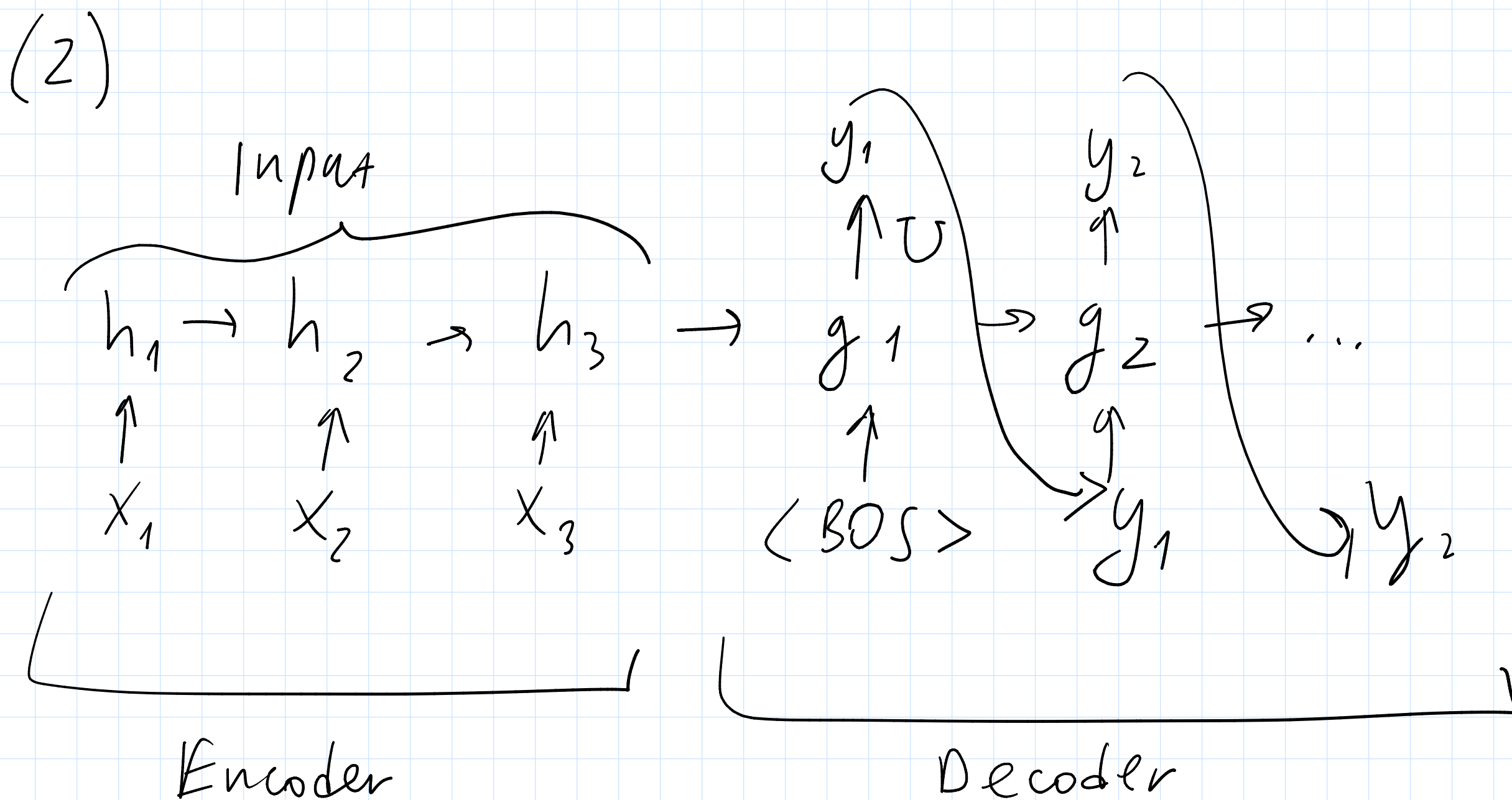
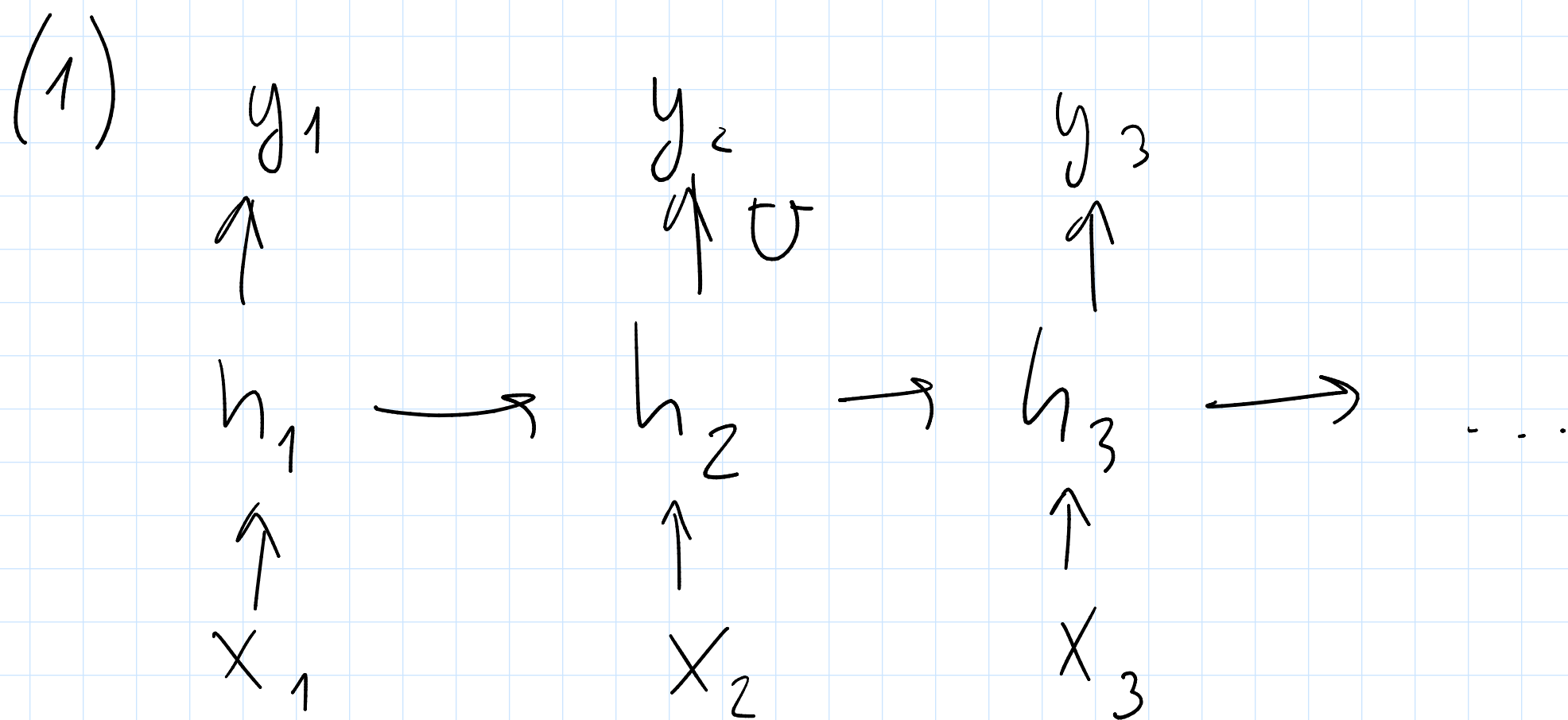
GRU (gated recurrent unit)



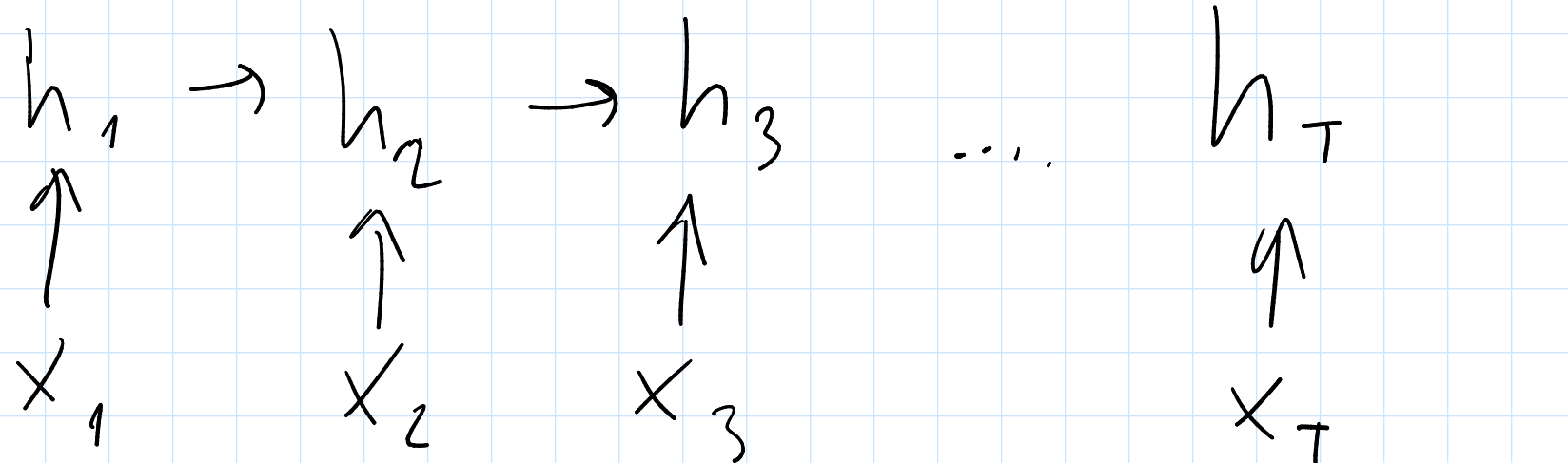
• Many-to-many (Seq-to-seq)

– Part of speech tagging (1)

– Machine translation (2)



- Many-to-one
 - Text classification

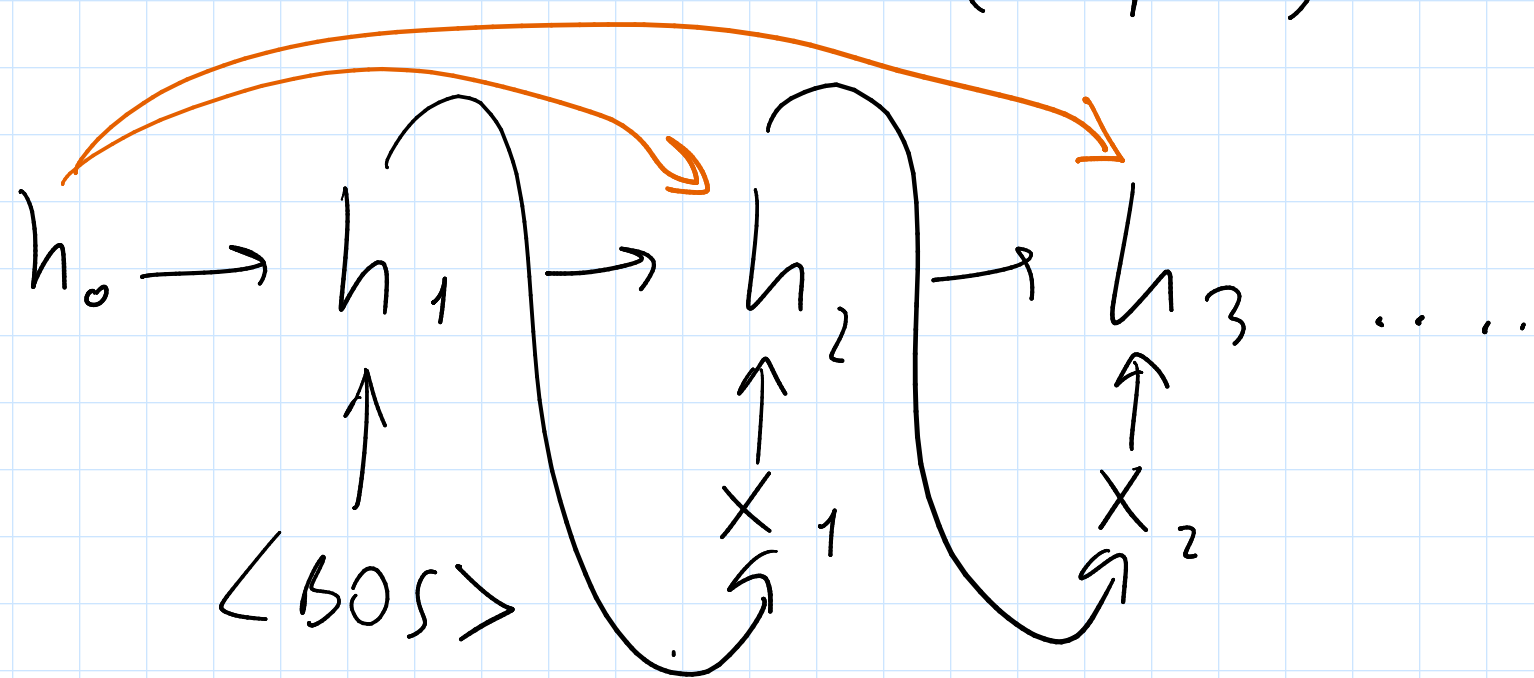


- One-to-many

- Conditional generation (Image captioning)

genre $\rightarrow$ music generation

$h_0 = \text{Encoder}(\text{Input})$ or $\text{Linear}(\text{Encoder}(\text{Input}))$



Bidirectional LSTM